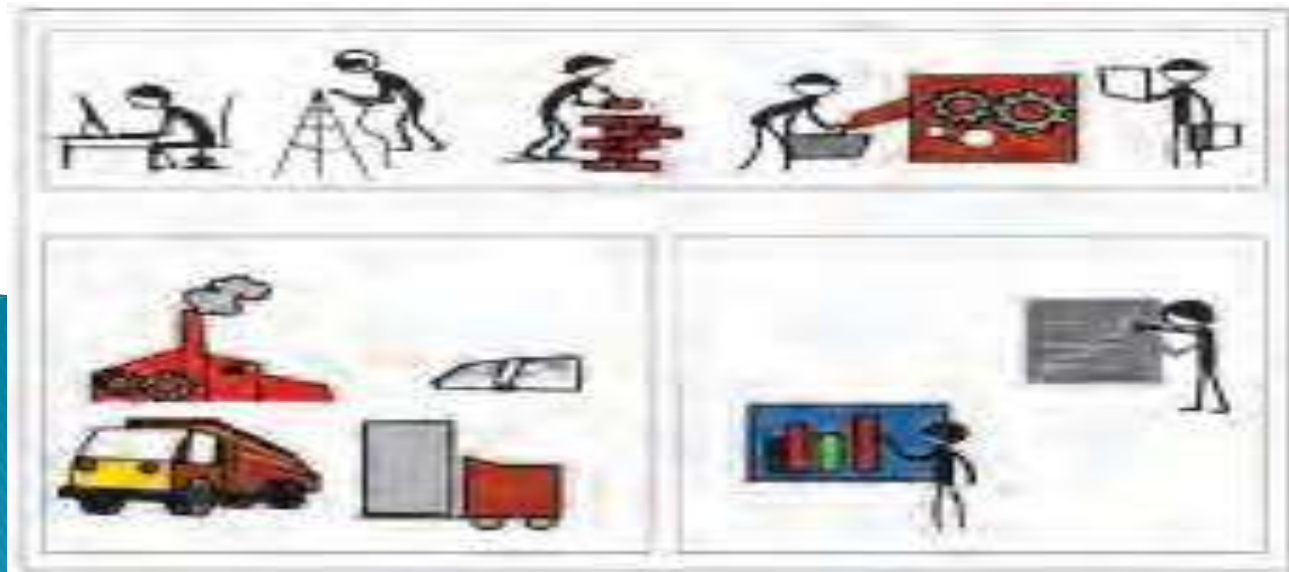


ERP

ENTERPRISE RESOURCE PLANNING



What is an Enterprise?

- 1 "An enterprise is a group of people and other resources working together for a common goal".
- 2 An enterprise acts as a single entity and an organization is divided into different units based on the operations performed in it.
- 3 An enterprise may consist of different sections such as manufacturing or production, planning, sales, purchase, finance, distribution etc.
- 4 Each department will have their own duties and responsibilities and they are working to achieve the objective which is set for the enterprise.
- 5 An enterprise can be considered as a **system** and all its departments as its **sub systems**.

Enterprise resource planning (ERP) refers to a type of software that organizations use to manage day-to-day business processes, such as accounting, project management, and supply chain operations. etc

ERP, is the process of gathering and managing business data from various departments using dedicated software.



Fig. 6.1 : An Enterprise

What is Resource?

There are different types of resources in an enterprise like **men, material, money and machine.**

Information system can be designed for various departments of an enterprise so that accurate and timely data can be provided to the concerned persons.

What is Planning?

Planning helps managers to improve future performance, by establishing objectives and selecting a course of action, for the benefit of the organization.



Fig. 6.2 : An enterprise with no or little Communication between departments

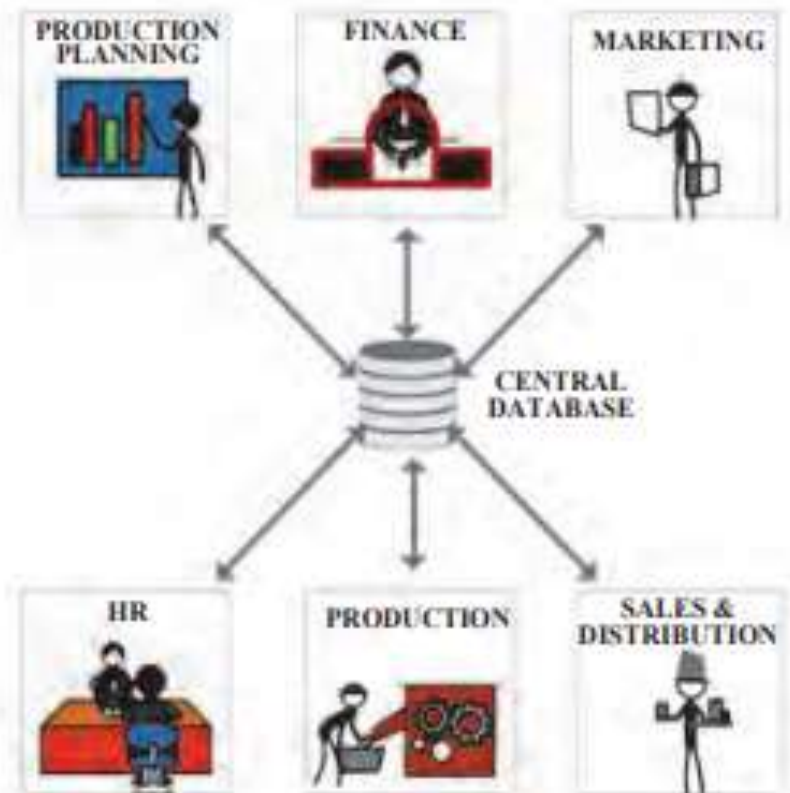
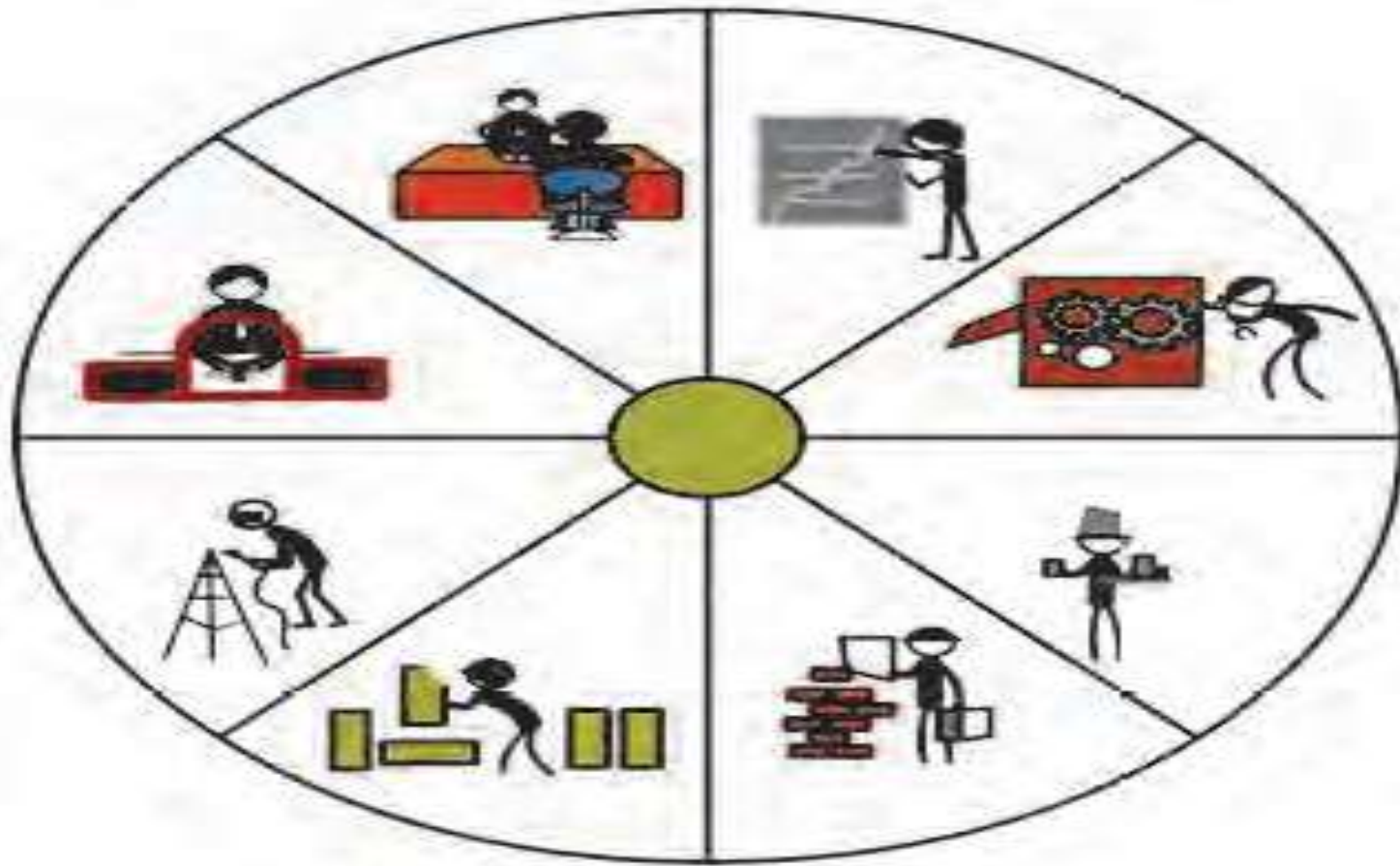


Fig. 6.3 : An enterprise with Central Database

Functional Units of ERP



Benefits of ERP

- ▶ 1. Improved resource utilization
- ▶ 2. Better customer satisfaction
- ▶ 3. Provides accurate information :
- ▶ 4. Decision making capability
- ▶ 5. Increased flexibility :
- ▶ 6. Information integrity :

Financial module

- 1] **This module** is the core of many ERP software packages.
- 2] this module allows businesses to understand their current financial state and future outlook.
- 3] It can collect financial data from various functional departments and generate valuable financial reports.
- 4] Financial reports include account payable , account receivable , balance sheets, general ledger, trial balance, financial statements, etc.
- 5] This module also includes financial accounting, investment management, enterprise controlling and treasury

Manufacturing module

- 1] Manufacturing module contains necessary business rules to manage the entire production process.
- 2] This module of ERP enables an enterprise to combine technology and business processes to get integrated solutions.
- 3] It also provides freedom to change manufacturing and planning methods as and when required.

Production planning module

- 1] This module is a critical component of ERP software for manufacturers.
- 2] It is used to plan and control the manufacturing process, from product design through production and delivery to customer.
- 3] This module is used for optimising the utilization of available resources and helps the organization to plan their production.
- 4] This module identifies the materials required, allocates optimal resources using data.

HR module

HR stands for Human Resource. HR module maintains an updated and complete employee database including personal information, salary details, attendance, performance, promotion, etc. of all employees in an enterprise.

Quality management module :

This module is used for managing the quality of the product. The quality management module fulfills the following functions–Quality planning, Quality inspection and Quality control.

Inventory control module :

This module covers processes of maintaining the appropriate level of stock in the warehouse. It is responsible for identifying the inventory requirements and setting the target of the stock items required.

6. Purchasing module :

Purchase Module helps for generating purchase order evaluating the supplier, and billing. It is closely connected with the inventory, finance and production planning module.

Marketing module :

Marketing module is used for monitoring and tracking customer orders, increasing customer satisfaction and for eliminating credit risks.

Sales and distribution module :

This module helps for tracking enquiries, order placement, order scheduling, dispatching and invoicing. This module is closely integrated with the e-commerce website of the organization.

ERP and related technologies

- ▶ **Product Life Cycle Management (PLM)**
- ▶ **Management Information System (MIS)**
- ▶ **Supply Chain Management (SCM):**
- ▶ **Customer Relationship Management(CRM) :**
- ▶ **Decision Support System (DSS) :**

ERP solution providers/ERP packages

- ▶ Oracle
- ▶ SAP
- ▶ Odoo
- ▶ Bitrix24
- ▶ Microsoft Dynamics
- ▶ Tally ERP

Limitations in ERP implementation

- ▶ **High cost**
- ▶ **Requirement of additional trained**
- ▶ **Staff**
- ▶ **Operational and maintenance**
- ▶ **Issues**
- ▶ **Security Control**